A popular IV

Draft lottery (1969)

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On December 1, 1969, the Selective Service System of the United States conducted two lotteries to determine the order of call to military service in the Vietnam War in the year 1970, for men born from January 1, 1944 to December 31, 1950. These lotteries occurred during a period of conscription in the United States that lasted from 1947 to 1973. It was the first time a lottery system had been used to select men for military service since 1942. The lottery would establish the priority of call based on the birth dates of registrants.

Computing complier characteristics

Useful special cases:

- $g(X_i, Y_i) = X_i$: complier mean of X_i
- $g(X_i, Y_i) = Y_i$: marginal complier means of $Y_i(1)$ and $Y_i(0)$
- $g(X_i, Y_i) = 1\{Y_i \quad y\}$: complier CDFs of $Y_i(1)$ and $Y_i(0)$ evaluated at y

This last result shows that the full marginal distributions of $Y_i(1)$ and $Y_i(0)$ are identified for compliers.

Back to overid tests

To fix ideas, return to a model with binary treatment T_i, binary instrument Z_i, and covariates X_i. Saturated 2SLS model:

$$Y_i = X_i'\alpha + \rho T_i + \varepsilon_i$$
$$T_i = X_i'\lambda + Z_i X_i'\pi + \eta_i$$

The overid test for this model checks whether Wald estimates are the same for each value of X_i.
Covariate-specific reduced forms and first stages:

$$d(x) = E[Y_i | Z_i = 1, X_i = x] \quad E[Y_i | Z_i = 0, X_i = x]$$
$$p(x) = E[T_i | Z_i = 1, X_i = x] \quad E[T_i | Z_i = 0, X_i = x]$$

Suppose the overid test rejects, and consider two possible plots: $\hat{d}(x)$ against $\hat{p}(x)$.

APPENDIX

Original many/weak slides, kept for comparison with the rewritten section. Also contains the trace derivation of the Bekker approximation.

The Judges Design

Common approach to the judges design: Define

$$T_i^* = \frac{1}{n_{j(i)} - 1} \sum_{k \neq i} 1\{j(k) = j(i)\} T_k$$

T_i^* is the mean treatment rate (e.g. incarceration rate) for the judge assigned to i, $j(i)$, for all cases not involving i

T_i^* is then used as an instrument for i's actual treatment T_i

The Judges Design: JIVE

Note that T_i^* is the fitted value from a first stage regression of removal on all the judge dummies, leaving out i

The instruments are implicitly a full set of judge dummies, which may lead to many/weak problems

Using a leave-out mean changes the estimator from 2SLS to jackknife IV (JIVE), which is one solution to the many/weak problem but not necessarily the optimal one

Conventional standard errors for JIVE are also incorrect in many/weak scenarios.

Despite this, many papers use judges design given its implicit attractiveness plus the possibility to summarize nicely heterogenous treatment effects (see the marginal treatment effects literature pioneered by Heckman and Vytacil, 1999,2005).